KAVIYA N

Python Full Stack Developer
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PROFESSIONAL SUMMARY

Python Full Stack Developer with 4 years of experience building scalable web applications, RESTful APIs, and production backend systems using Django, Django REST Framework, FastAPI, React.js, Laravel, and PHP. Strong expertise in PostgreSQL/MySQL design, query optimization, Docker deployment, JWT/OAuth security, CI/CD, and Agile/Scrum delivery. Exposure to AI/ML solutions including Computer Vision, Generative AI, FAISS vector search, and 3D reconstruction workflows.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, PHP
Frameworks & Libraries: Django, Django REST Framework (DRF), FastAPI, Flask, React.js, Laravel
API Development: RESTful APIs, API Integration, API Design, API Documentation
Databases: PostgreSQL, MySQL, SQLite, SQL
Cloud & DevOps: Docker, Kubernetes, CI/CD Pipelines, AWS, Microsoft Azure, Linux, Git, GitHub
Security & Testing: JWT, Authentication & Authorization, API Security, PyTest, Unit Testing, API Testing, Postman
AI / ML: Machine Learning, Deep Learning, Computer Vision, Generative AI, Prompt Engineering, Embeddings, Vector Search, FAISS, Retrieval-Augmented Generation (RAG)
Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Software Development Life Cycle (SDLC), Agile/Scrum Methodology

PROFESSIONAL EXPERIENCE

Backend Developer | Binary Bot | January 2025 - Present

- Engineered and maintained scalable server-side components using Python, Django, Django REST Framework, FastAPI, PostgreSQL, Docker, and Linux across multiple workflows.
- Designed RESTful APIs for product search, image upload, onboarding, authentication, workflow automation, and 3D visualization.
- Built an AI-powered Product Visual Search Engine using OpenCV, DINOv2, and FAISS-backed similarity search.
- Optimized response times through PostgreSQL query optimization, indexing, caching, and backend performance tuning for high-traffic endpoints.
- Implemented JWT authentication, OAuth aligned security, role-based workflows, input validation, and secure API endpoints.
- Automated CI/CD and Docker deployment workflows while handling migrations, debugging, testing, code reviews, and production support.

Backend Developer | Yaane Technologies | September 2022 - December 2024

- Supported live production healthcare and patient safety management systems using Laravel, PHP, MySQL, React.js, and REST APIs.
- Tuned MySQL queries, joins, and database logic to improve reliability, application performance, and maintainability.
- Strengthened secure authentication workflows, validation rules, role-based access control, and business process automation.
- Integrated API layer with React.js components and collaborated with frontend teams to deliver stable full-stack features across live releases.
- Led production debugging, API testing, deployment activities, release support,Agile/Scrum , SDLC tasks, and issue resolution.

PROJECTS

AI Product Visual Search Engine

Tech Stack: Python, Django, OpenCV, DINOv2, FAISS, PostgreSQL, Docker, REST APIs, JWT

- Engineered an AI-powered visual search system that allows users to upload product images and retrieve visually similar products.
- Delivered RESTful APIs for image upload, search workflow, authentication, and result delivery using Django and Django REST Framework.
- Applied image preprocessing and feature extraction using OpenCV and DINOv2 to support similarity-based search.
- Used FAISS-backed nearest-neighbor search to improve matching performance.

3D Product Viewer from Photos

Tech Stack: FastAPI, OpenCV, COLMAP, NeRF, Instant-NGP, Three.js, WebGL, PostgreSQL, Docker

- Designed processing services to transform product images and support a browser-based 3D product visualization workflow.
- Exposed FastAPI endpoints for image upload, camera pose estimation, model generation, and workflow orchestration.
- Integrated computer vision and 3D reconstruction tools including OpenCV, COLMAP, NeRF, and Instant-NGP.
- Connected backend APIs with a Three.js and WebGL viewer for interactive 3D product display in the browser.

Healthcare and Patient Safety Management System

Tech Stack: Laravel, PHP, MySQL, React.js, REST APIs

- Delivered secure backend modules for patient-record management, document review, approval flows, notifications, and reporting.
- Constructed reusable REST APIs and business logic to support live production workflows in a healthcare environment.
- Enhanced database performance using query tuning, joins, indexing, and structured relational schema design for reliable production use.
- Integrated backend services with React.js screens and supported end-to-end feature delivery.

EDUCATION

Bachelor of Engineering in Computer Science, Coimbatore Institute of Technology, Coimbatore, Tamil Nadu | Graduated: 2021

CERTIFICATIONS

Smart India Hackathon 2020

- Selected participant in institute-level internal hackathon.
- Contributed to solution design and presentation.
- Applied teamwork, problem solving, rapid prototyping, and software development practices under time constraints.